The University of Azad Jammu and Kashmir

Subject:

Software Design & Architecture

Roll Number:

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Name:

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Lab Title:

Food delivery system

Title: Food Delivery System

Task 1: List 6–10 Candidate Conceptual Classes:

- Customer – The person who orders food.
- Restaurant – The place that prepares food.
- Food Item – The food that a customer can order.
- Order – The record of food ordered by a customer.
- Delivery Rider – The person who delivers the food
- Payment – The record of money paid for an order.
- Address – The location where food is delivered.
- Menu – The list of food items available at a restaurant.
- Delivery – The process of delivering an order.
- Review – The feedback given by a customer.

These classes represent the main objects and concepts of a Food Delivery System.

Task 2: Attributes and Methods of Three Important Classes

- Customer Class:

Attributes:

Customer ID

Name

Phone number

Address

Methods:

placeOrder() – Places a food order.

cancelOrder() – Cancels an order.

giveReview() – Gives feedback about food or service.

- Order Class:

Attributes:

Order ID

Order Date

Total Amount

Order Status

Methods:

addFoodItem() – Adds food to the order.

calculateTotal() – Calculates the total bill.

updateStatus() – Changes the order status.

- Delivery Rider Class:

Attributes:

Rider ID

Name

Phone Number

Availability Status

Methods:

acceptDelivery() – Accepts a delivery request.

pickUpOrder() – Collects food from the restaurant.

deliverOrder() – Delivers food to the customer.

Task 3: Abstraction Decisions:

1.We include important details such as customer name, address, and order details.

2. We ignore unnecessary details like the customer's height, eye color, and favorite clothes.

3. We ignore the rider's personal life because it is not related to food delivery.

4. This makes the system simple, clear, and easy to use.

Task 4: Identify One Invented Software Object:

Invented Object: Order Controller

An Order Controller is a software object used to manage the order process. It does not exist as a physical object in the real world. It helps the system receive orders, check order details, and send orders to the restaurant. It also connects different parts of the software and helps the Food Delivery System work properly.